

Twisted Black p-Brane Solutions in String Theory

Ashoke Sen

Abstract

It has been shown that given a classical background in string theory which is independent of x^μ of the space-time coordinates, we can generate other classical backgrounds by $O(d) \times O(d)$ transformation on the solution. We study the effect of this transformation on the known black p-brane solutions in string theory, and show how these transformations produce new classical solutions labelled by extra continuous parameters and containing background antisymmetric tensor field.

1 Introduction

It has been shown that given a classical background in string theory which is independent of x^μ expression of the space-time coordinates, we can generate other classical backgrounds by $O(d) \times O(d)$ transformation on the solution. We study the effect of this transformation on the known black p -brane solutions in string theory, and show how these transformations produce new classical solutions labelled by extra continuous parameters and containing background antisymmetric tensor field.

2 Method

$g_{\mu\nu}$ where $g_{\mu\nu}$, ϕ and $B_{\mu\nu}$ denote the graviton, the dilaton, and the antisymmetric tensor fields respectively, α + cyclic permutations, R denotes the d dimensional Ricci scalar, and Λ is the cosmological constant equal to Λ_B for bosonic string and Λ_F for fermionic string. (For simplicity we have set the other background massless fields, which appear in fermionic string theories, to zero.) Let us now split the coordinates x^μ into two sets x^μ and y^μ (x^μ , y^μ) and consider backgrounds independent of x^μ . Let us further concentrate on backgrounds where ϕ , i.e. to backgrounds of the form ϕ , $B_{\mu\nu}$. In this case, after an integration by parts, the action S can be shown to take the form: $S = \int d^d x \int d^p y \sqrt{-g} \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} \partial_\mu B_{\nu\rho} \partial^\mu B^{\nu\rho} - \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} \partial_\mu B_{\nu\rho} \partial^\mu B^{\nu\rho} \right)$

$$L(\theta) = \sum_{i=1}^n \ell(x_i, \theta) \quad (1)$$

Equation 1 is included to keep the sample paper-like while remaining small.

3 Conclusion

If the coordinates x^μ are all of Euclidean signature, this action is invariant under an $O(d) \times O(d)$ transformation on x^μ , y^μ and $B_{\mu\nu}$, given by, $M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$